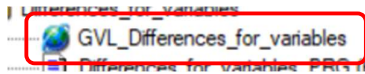


Properties for variables.

When declaring a variable there are 4 options to add to a variable.

1. Nothing
2. CONSTANT
3. RETAIN
4. PERSISTENT



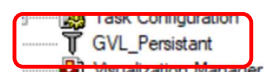
Explanation of the options:

1. When added nothing to a variable the value of the variable will be set to the default* value on almost every action done with the PLC.
2. When added "CONSTANT" to the variable the variable only can be modified by change it in the source program, this means you must be off-line and modify it in the software code.
3. When added "RETAIN" to the variable the variable keeps its value also when the power is being switched off and after a "Warm reset". The value will be set to the default* value after a "Reset Origin", and "Reset Cold" and after repeated program download.
4. When added "PERSISTANT" to the variable "RETAIN" also must be added to the variable. When this is done a special global variable list will be added to the system.

```
VAR_GLOBAL
byVar00: BYTE;
byVar01: BYTE;
byVar02: BYTE;
byVar03: BYTE;
// Normal variable
byTest_00: BYTE;
END_VAR

VAR_GLOBAL CONSTANT
// CONSTANT
byTest_01: BYTE := 25;
END_VAR
```

```
VAR_GLOBAL RETAIN
// RETAIN
byTest_02: BYTE;
END_VAR
```



The value of the variable will be kept on all time, this means that it will be kept after: a reload of the program, after a download of the program, after a "Reset Warm" and even after a "Reset Cold!"

The only way to reset it to default* is after a "Reset Origin", but the PLC is then reset to the factory state, the PLC is completely empty, included the program!

```
GVL_Persistent
1
2 VAR_GLOBAL PERSISTENT RETAIN
3 // PERSISTENT must also be declared as RETAIN
4 byTest_03: BYTE;
5 END_VAR
```

*Set to "Default" means: the value will be set to 0 or to the Initialization value.

```
byTest_00: BYTE;    = set to "0" (or "FALSE" for a BOOL)
byTest_01: BYTE := 25;    = set to "25"
```